

## David Sebastian Broska

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### Education

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|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2021 - Present | <b>Stanford University</b><br>PhD Sociology, expected 2027<br>Dissertation: Generative AI for the Social Sciences<br>Committee: Robb Willer, David Grusky, Jeremy Freese, Michelle Jackson |
| 09/2019 - 08/2020 | <b>London School of Economics and Political Science (LSE)</b><br>MSc Social Research Methods                                                                                               |
| 09/2014 - 07/2019 | <b>Zeppelin University (ZU), Friedrichshafen</b><br>BA Sociology, Politics & Economics                                                                                                     |

### Honors and Fellowships

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|-------------------|-------------------------------------------------------------------------------|
| 02/2025           | Best Prediction Algorithm Award, <a href="#">Wisdom of the Crowds Project</a> |
| 09/2024           | Best Poster Award, Digitalist Papers Symposium, Stanford University           |
| 05/2024           | Centennial Teaching Assistant Award, Stanford University                      |
| 05/2024 - 08/2024 | Research Computing Graduate Fellowship, Stanford University                   |
| 09/2023 - 06/2024 | Human-Centered AI Graduate Fellowship, Stanford University                    |
| 11/2020           | Prize for the Best Overall Performance in MSc Research Methods, LSE           |

### Peer-Reviewed Publications

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| 2026 | <b>D. Broska</b> , J. Kamphorst, and R. Willer. Americans Favor Predistributive Over Redistributive Economic Policies. Forthcoming at <a href="#">PNAS</a> .                                           |
| 2025 | <b>D. Broska</b> , M. Howes, and A. van Loon. The Mixed Subjects Design: Treating Large Language Models as Potentially Informative Observations. <a href="#">Sociological Methods &amp; Research</a> . |
| 2024 | D. McFarland, <b>D. Broska</b> , V. Prabhakaran, and D. Jurafsky. Coming into Relations: How Communication Reveals and Persuades Relational Decisions. <a href="#">Social Networks</a> .               |

### Work in Progress

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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | M. Binz, ..., <b>D. Broska</b> , ... Post-training Makes Large Language Models Less Human-like. <a href="#">arXiv preprint</a> .                                                              |
| 2026 | J. Hullman, <b>D. Broska</b> , H. Sun, and A. Shaw. This Human Study Did Not Involve Human Subjects: Validating LLM Simulations as Behavioral Evidence. <a href="#">arXiv preprint</a> .      |
| 2026 | <b>D. Broska</b> , L. Santos, and R. Willer. Belief in the American Dream Reduces Support for Welfare Programs. <i>Manuscript available upon request</i> .                                    |
| 2026 | <b>D. Broska</b> , B. Park, A. Mack-Crane, D. McFarland, and B. Lee. Civility Is Necessary but Not Sufficient for Productive Conversations Online. <i>Manuscript available upon request</i> . |

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|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>D. Broska</b> , L. Hewitt, ..., R. Willer. Predicting Experimental Effect Sizes with Finetuned LLMs. <i>Manuscript available upon request.</i>                                                |
| 2026 | <b>D. Broska</b> and M. Howes. Cluster-Robust Prediction-Powered Inference. <i>Manuscript available upon request.</i>                                                                            |
| 2026 | N. Stagnaro, <b>D. Broska</b> , and H. Lin. Behavioral traces distinguish bots and AI-assisted humans in online surveys. <i>Manuscript available upon request.</i>                               |
| 2026 | J. Kamphorst, <b>D. Broska</b> , M. Bernstein, and R. Willer. Leveraging Large Language Models to Predict Results of Social Science Field Experiments. <i>Manuscript available upon request.</i> |

## Selected Presentations

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|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 06/2026 | <b>Shaping the Future of AI-Enabled Surveys, University of Mannheim</b><br>Paper presentation: Fine-tuning LLMs to Forecast Experimental Results               |
| 05/2026 | <b>Sociological Science Conference, Stanford University</b><br>Paper presentation: The Mixed Subjects Design                                                   |
| 05/2026 | <b>FDS Workshop: AI for Social Science Research Methods, Yale University</b><br>Paper presentation: Fine-tuning LLMs to Forecast Experimental Results          |
| 02/2026 | <b>Symposium on Validating Generative AI, Northwestern University</b><br>Paper presentation: The Mixed Subjects Design                                         |
| 01/2026 | <b>Nuffield College Sociology Seminar, University of Oxford</b><br>Invited talk: AI for the Experimental Social Sciences                                       |
| 08/2025 | <b>Adaptive Rationality Seminar, Max Planck Institute Berlin</b><br>Invited talk: AI for the Experimental Social Sciences                                      |
| 07/2025 | <b>Infer Mittelbau Seminar, Goethe University</b><br>Paper presentation: Belief in the American Dream Reduces Support for Welfare Programs                     |
| 04/2024 | <b>Generative AI for Sociology Workshop, Yale University</b><br>Paper presentation: The Mixed Subjects Design                                                  |
| 08/2023 | <b>American Sociological Association Annual Meeting, Philadelphia</b><br>Paper presentation: Belief in the American Dream Reduces Support for Welfare Programs |

## Teaching

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|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 09/2023 - 12/2023                       | <b>Economic Sociology (SOC 214), Stanford University</b><br>Teaching Assistant; course on the sociological analysis of markets                     |
| 04/2023 - 06/2023;<br>04/2025 - 06/2025 | <b>Regression for Discrete Outcomes (SOC 383), Stanford University</b><br>Teaching Assistant; regression analysis for discrete dependent variables |
| 02/2021 - 05/2021                       | <b>Computational Social Science (243113), Zeppelin University</b><br>Lecturer; computational text analysis                                         |
| 02/2021 - 05/2021                       | <b>Empirical Research Design (10007), Zeppelin University</b><br>Lecturer; empirical research design in the social sciences                        |

## Research Funding and Grants

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|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 08/2026 | <b>Stanford Impact Labs (SIL): \$25,000</b><br>Predicting Experimental Effect Sizes with Fine-tuned LLMs.<br>Role: Project lead; PI: Robb Willer.                                                                          |
| 03/2026 | <b>Stanford Institute for Human-Centered Artificial Intelligence (HAI): \$75,000</b><br>Leveraging Large Language Models to Predict Results of Social Science Field Experiments. Role: Project co-author; PI: Robb Willer. |
| 12/2025 | <b>King Center on Global Development, Stanford University: \$15,000</b><br>Leveraging Large Language Models to Predict Results of Global Development Field Experiments. Role: Project co-author; PI: Robb Willer.          |
| 04/2024 | <b>HAI-Azure Cloud Credits Award, Stanford University: \$500</b><br>The Mixed Subjects Design. Treating Large Language Models as Potentially Informative Observations. Role: Project lead.                                 |

## Professional Experience

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|-------------------|---------------------------------------------------------------------------------------------------|
| 09/2020 - 08/2021 | <b>Zeppelin University, Friedrichshafen</b><br>Research and Teaching Fellow; research methodology |
| 03/2018 - 07/2018 | <b>EUROSTAT at European Commission, Luxembourg</b><br>Trainee; programming and data engineering   |
| 09/2017 - 01/2019 | <b>Campusjäger Recruiting GmbH, Karlsruhe</b><br>Working Student; data analysis in R and MySQL    |

## Service

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 01/2022 - Present | <b>Buddy Program, Stanford University</b><br>Mentor to first-year PhD students                                              |
| 09/2022 - 08/2024 | <b>Association of Graduate Students, Stanford University</b><br>Social Events Chair for Sociology graduate students         |
| 01/2016 - 06/2016 | <b>Buddy Initiative, Zeppelin University</b><br>Mentor to first-year undergraduate students                                 |
| 09/2013 - 05/2014 | <b>Lycée des Métiers Sonnenberg de Carspach, Mulhouse</b><br>Volunteer and language assistant in a French vocational school |

## Reviews

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|-----------------|----------------------------------------------------------------------|
| Journals        | Political Analysis, Sociological Methods & Research, Network Science |
| Grant proposals | National Science Centre (NCN), Poland                                |

## Languages and Computer Programming

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|-------------|-------------------------|
| Languages   | German, English, French |
| Programming | R, SQL, Git, Python     |